Counting positive rational friezes over $\frac{1}{N}\mathbb{Z}$

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Abstract

Let $T(N,n)$ denote the number of positive rational frieze patterns of width n all of whose entries lie in $\frac{1}{N}\mathbb{Z}$. Conway and Coxeter showed $T(1,n) = C_{n-2}$. We prove that for every prime $p \geq 5$

$$T(p,5) = 5 + 5C(p), \qquad C(p) = \#\{(a,u,v) \in \mathbb{Z}^3_{>0} : auv = p + u + v\},$$

an equality identifying the width-five count with the case $A = 1$, $B = p$ of $xyz = A(x+y)+B$, a problem posed by Ford in generalisation of a question of Conrey. The order of $C(p)$ is open, so no closed form for $T(p,5)$ should be expected.

These friezes are classified in [5] by the minimal clockwise paths of a Farey graph. That classification is counted here: the number of minimal closed clockwise paths of length five in $\mathcal{F}_R$, modulo $SL_2(\mathbb{Z})$, is $\sum_{d|R} \mu(R/d)T(d,5)$, and at a prime it is $5C(p)$.

At every N the count decomposes the same way. It is a finite sum of counts for Ford's equation, one term for each divisor M of N^2 with $N \mid M^2$, the term carrying the parameters $A = \gcd(N,M)$ and $B = M\gcd(N,M)/N$. The map $M \mapsto M + N$ identifies that index set with the reflection-symmetric friezes, so the number of terms is $Q(N)$, the number of square divisors of N^3, and $T(N,5) \equiv 5Q(N) \bmod 10$. At a prime the sum has two terms and collapses to the display above; for composite N it does not collapse, and no single instance of Ford's equation carries the count. Sorting each term by its least variable, which (8) bounds, improves the general upper bound to $T(N,5) = O_\varepsilon(N^{2/3+\varepsilon})$. The parameters that occur are determined exactly: a pair (A,B) arises if and only if $B \mid \gcd(A,B)^2$. So the conjecture $T(N,5) = N^{o(1)}$ is not merely implied by Ford's, it is equivalent to Ford's restricted to that family. The transfer also runs backwards: the free rotation action forces 5 to divide the sum of Ford counts, although it divides almost none of the summands.

We also prove $T(N,4) = d(2N^2)$, give proved search bounds at widths 5 and 6, so that the tabulated values are complete rather than observed to stabilise, and tabulate $T(N,n)$ for widths 3 to 6 and, at small N, for widths 8 and 9. The rotation action of $\mathbb{Z}/5$ on width-5 quiddity cycles is free, so $5 \mid T(N,5)$. We prove $T(N,5) = O_\varepsilon(N^{2/3+\varepsilon})$, and $T(p,5) = O_\varepsilon(p^{1/3+\varepsilon})$ at primes by importing the method of Conrey and Shah, while $T(p,5) \geq 5\,d(p+1)$ refutes the guess $T(N,5) \asymp d(N^2)\log^2 N$.

The reduction behind the tables holds at every width: homogenising the frieze continuant and applying the Desnanot–Jacobi identity collapses the lattice conditions to $gUV = N^{n-4}(U+V)+M$ with $M \mid N^2R$ and $N^{n-4}R \mid gM$, where R is a continuant window empty at width 5.

The count at a prime is machine-checked in Lean 4 from the definition of a frieze itself, with periodicity derived rather than assumed; Section 13 says what else is verified and what is not.

1 Introduction

A frieze pattern of width n is a finite array of rationals bordered by zeros and ones in which every adjacent 2×2 minor equals 1. Conway and Coxeter [4] classified the patterns with positive integer entries: they correspond to triangulations of a convex n-gon, so there are C_{n-2} of them. Allowing denominators changes the problem. Fixing N and asking for the positive patterns with all entries in $\frac{1}{N}\mathbb{Z}$ gives a count $T(N, n)$, finite because $\frac{1}{N}\mathbb{Z}$ is discrete and Cuntz and Holm [3] showed a discrete subset admits only finitely many friezes of each height, and computing it is the subject of this paper.

The point of this paper is that this count is an open problem in analytic number theory wearing different clothes. At a prime,

$$T(p, 5) = 5 + 5C(p), \qquad C(p) = \#\{(a, u, v) \in \mathbb{Z}^3_{>0} : auv = p + u + v\}$$

(Theorem 7.7). The width-5 reduction is $guv = N(u + v) + M$, which is $xyz = A(x + y) + B$ with $A = N$ and $B = M$, posed by Ford in generalisation of a question of Conrey and recorded in [9]; the cubic $auv = p + u + v$ above is its case $A = 1$, $B = p$. Ford appears to have published nothing further on it. The fully symmetric variant $n = xyz + x + y + z$, also asked by Conrey, is Problem 25 of the American Institute of Mathematics problem list of May 2026 and is the equation of Conrey and Shah [7]. The order of $C(p)$ is not known. Counting friezes over $\frac{1}{p}\mathbb{Z}$ and estimating $C(p)$ are the same problem.

Friezes have been related to equations of this shape before, and the difference matters. Zhang [1] relates friezes to $xyz = G(x, y)$ in a different setting and direction: positive *integral* friezes of Dynkin type, where the Fontaine–Plamondon counts are determined exactly, and where the Diophantine conclusion is the existence of infinitely many positive solutions rather than a bound on their number. The equations here are the rational ones over $\frac{1}{N}\mathbb{Z}$, and the question is the counting bound asked by Conrey and, in generalised form, by Ford. Conley and Ovsienko [2] enumerate the positive *integer* solutions of the same matrix equation, graded by a parameter measuring the departure of a dissection from a triangulation; that parameter and the denominator N are independent, and none of the sequences below occurs in their tables.

Theorem 7.7 is an equality, not an estimate, so an asymptotic for either side transfers to the other. Neither side is settled here. The count is $O_\varepsilon(p^{1/3+\varepsilon})$ by the method of Conrey and Shah [7] (Theorem 9.3), against a conjectured $p^{o(1)}$ (Conjecture 9.5). The lower bounds of Propositions 9.6 and 9.7, $T(p, 5) \geq 5\, d(p + 1)$ and $C(p) \geq d(p + 1) + d(2p + 1)$, are the $a = 1$ and $a = 2$ slices; Conrey and Shah record the same two for their equation in [7, §2.1]. Proposition 9.8 puts the count on the modular hyperbola $de \equiv 1 \bmod p$, where the Weil error is of size $p^{1/2}$ against a quantity of size $p^{1/3}$, which is why the standard route does not reach even the trivial bound.

For frieze theory this explains an absence. $T(N, 4) = d(2N^2)$ has a closed form, counts of friezes over $\mathbb{Z}/n\mathbb{Z}$ are multiplicative with explicit formulas, and no such formula has been found at width 5 over $\frac{1}{N}\mathbb{Z}$: finding one would settle a question of Conrey. The friezes counted here are exactly those classified in [5]. In the other direction it gives $C(p)$ a combinatorial reading, as the count of a set of friezes, which is not how the quantity in Ford's problem arises elsewhere.

Three further reductions are carried out below. Section 10 gives the width-six analogue of the width-five reduction: a master identity $AB = N^4 + N^2qf$ of Ford type and a system of divisibilities in four variables that is necessary and sufficient, where the defining conditions alone are not. Section 11 observes that the width-five reduction uses no order, so it holds over any commutative domain; over $\mathbb{Z}[i]$ the solution set is still finite, by a norm bound in place of the search bounds of the real theory, and two nondegeneracy conditions that positivity hides must be imposed. Section 12 makes the passage to the Farey graph explicit rather than numerical: the frieze relation is the Plücker relation in the plane, which gives the correspondence at every width at once, and positivity of a frieze is exactly the clockwise order of the path.

Section 7 tabulates $T(N,n)$ for widths 3 to 6 and for widths 8 and 9 at small N. At width 4 the count has a closed form, $T(N,4) = d(2N^2)$ (Theorem 4.1). The reductions and search bounds behind the tables at the larger widths are routine and are carried out in the accompanying library rather than here.

Section 13 says which results are machine-checked and which rest on a written proof; the per-statement map to Lean declarations is maintained with the library. Tables and computed constants are regenerated by scripts in `code/`, named at the point of use; `code/paper_numbers.py` collects those that have no script of their own.

2 Setting

We use the conventions of [5].

Definition 2.1. A *rational frieze of width* n is a map $F\colon \{(i,j) : 0 \le i-j \le n\} \to \mathbb{Q}$, written $m_{i,j}$, with $m_{i,j} = 0$ when $i-j \in \{0,n\}$, with $m_{i,j} = 1$ when $i-j \in \{1, n-1\}$, and satisfying the diamond rule

$$m_{i,j}\, m_{i+1,j+1} - m_{i,j+1}\, m_{i+1,j} = 1.$$

It is *positive* if every entry with $0 < i-j < n$ is positive. The *quiddity cycle* is the row $i-j=2$, a sequence $(a_0,\ldots,a_{n-1})$ read cyclically.

The quiddity cycle determines the frieze: writing r for the row index,

$$m_{r,j} = \frac{m_{r-1,j}\, m_{r-1,j+1} - 1}{m_{r-2,j+1}}, \tag{1}$$

with the closing conditions that row $n-1$ is constant 1 and row n is constant 0.

Definition 2.2. For $N \ge 1$ let $T(N,n)$ be the number of positive rational friezes of width n all of whose entries lie in $\frac{1}{N}\mathbb{Z}$. Quiddity cycles are counted as sequences, not up to rotation.

Conway and Coxeter [4] showed that the positive *integer* friezes of width n correspond to triangulations of a convex n-gon, so

$$T(1,n) = C_{n-2} = \frac{1}{n-1}\binom{2n-4}{n-2}. \tag{2}$$

Finiteness of $T(N,n)$ is Corollary 3.8 of Cuntz and Holm [3], which gives it for any discrete $R \subseteq \mathbb{C}$. Theorem B of [5] classifies the positive rational friezes over $\frac{1}{N}\mathbb{Z}$ in terms of minimal clockwise paths in a Farey-type graph. It classifies exactly the objects counted here, and does not count them. The present note is concerned with computing it.

3 The continuant parameterisation

Let K_k denote the frieze continuant, $K_k = x_k K_{k-1} - K_{k-2}$ with $K_0 = 1$, $K_{-1} = 0$. Continuants and the continued-fraction geometry behind them are treated in [6].

Proposition 3.1. *A positive rational frieze of width $n \geq 4$ is determined by $a_0, \ldots, a_{n-4}$, and*

$$a_{n-3} = \frac{K_{n-4}(a_0, \ldots, a_{n-5}) + 1}{K_{n-3}(a_0, \ldots, a_{n-4})}, \qquad a_{n-2} = K_{n-3}(a_0, \ldots, a_{n-4}),$$
$$a_{n-1} = \frac{K_{n-4}(a_1, \ldots, a_{n-4}) + 1}{K_{n-3}(a_0, \ldots, a_{n-4})}.$$

Proof. The closing conditions are $m_{n-1,j} = 1$ and $m_{n,j} = 0$ for every j, that is $K_{n-2}(a_j, \ldots, a_{j+n-3}) = 1$ and $K_{n-1}(a_j, \ldots, a_{j+n-2}) = 0$. Expanding the second at $j = 0$ by the recursion $K_k = x_k K_{k-1} - K_{k-2}$ in its last argument and solving for a_{n-2} gives the middle display; expanding the first at $j = 0$ and at $j = 1$ and solving for a_{n-3} and a_{n-1} gives the other two. The remaining $n-3$ closing conditions are cyclic conjugates of these under the monodromy $M(a_0) \cdots M(a_{n-1}) = -I$ and impose nothing further. □

At $n = 5$ this reads

$$a_2 = \frac{a_0 + 1}{D}, \qquad a_3 = D, \qquad a_4 = \frac{a_1 + 1}{D}, \qquad D = a_0 a_1 - 1, \tag{3}$$

and at $n = 6$, with $D = K_3(a_0, a_1, a_2) = a_2(a_0 a_1 - 1) - a_0$,

$$a_3 = \frac{a_0 a_1}{D}, \qquad a_4 = D, \qquad a_5 = \frac{a_1 a_2}{D}. \tag{4}$$

Remark 3.2. A frieze of width n has a glide symmetry exchanging row r with row $n - r$. For $n = 5$ the only non-constant interior rows are 2 and 3, and they are exchanged, so a quiddity cycle in $\frac{1}{N}\mathbb{Z}$ forces every entry into $\frac{1}{N}\mathbb{Z}$. For $n = 6$ row 3 is fixed by the glide, and carries a lattice condition of its own. This is the source of the extra divisibility constraints at width 6.

4 Width four

Write d for the number-of-divisors function.

Theorem 4.1. *For every $N \geq 1$, $T(N, 4) = d(2N^2)$.*

Proof. Let (a_0, a_1, a_2, a_3) be the quiddity cycle. By (1) the row 3 entries are $a_j a_{j+1} - 1$, and row 3 is constant 1, so $a_j a_{j+1} = 2$ for all j modulo 4. Hence $a_0 = a_2$, $a_1 = a_3$ and $a_0 a_1 = 2$. Conversely any $a > 0$ with $a \cdot (2/a) = 2$ yields the frieze with quiddity $(a, 2/a, a, 2/a)$, whose remaining rows are forced and positive.

Write $a_0 = p/N$ with p a positive integer. Then $a_1 = 2N/p$, and $a_1 \in \frac{1}{N}\mathbb{Z}$ if and only if $2N^2/p \in \mathbb{Z}$, that is $p \mid 2N^2$. Distinct divisors give distinct cycles, so $T(N, 4)$ is the number of positive divisors of $2N^2$. □

The sequence $d(2N^2)$ is OEIS A361689.

5 Width five

Put $a_0 = p/N$ and $a_1 = q/N$ with p, q positive integers, and set $e := pq - N^2$.

Proposition 5.1. *The parameters (p, q) arise from a positive frieze of width 5 over $\frac{1}{N}\mathbb{Z}$ if and only if*

$$e > 0, \qquad N \mid e, \qquad e \mid N^2(p + N), \qquad e \mid N^2(q + N). \tag{5}$$

Proof. By (3), $a_3 = D = e/N^2$, $a_2 = N(p+N)/e$ and $a_4 = N(q+N)/e$. Positivity of the frieze is equivalent to $D > 0$, that is $e > 0$. Membership of a_3 in $\frac{1}{N}\mathbb{Z}$ is $N \mid e$, and membership of a_2 and a_4 is the two divisibility conditions. By Remark 3.2 no further condition is needed. □

Lemma 5.2 (Search bound)**.** *Any solution of* (5) *satisfies* $p \le N^3 + 2N^2$ *and* $q \le N^3 + 2N^2$.

Proof. Since $e > 0$ and $e \mid N^2(q+N)$ with $N^2(q+N) > 0$, we have $e \le N^2(q+N)$. Substituting $e = pq - N^2$ gives $pq - N^2 \le N^2 q + N^3$, that is $q(p - N^2) \le N^3 + N^2$. As $q \ge 1$, either $p \le N^2$ or $p - N^2 \le N^3 + N^2$; in both cases $p \le N^3 + 2N^2$. The bound on q is symmetric. □

Lemma 5.2 makes the enumeration finite with an explicit range. Since additionally $e \mid N^2(p+N)$ and $q = (e+N^2)/p$, one may loop over p in the proved range and over the divisors e of $N^2(p+N)$, which is what `code/verify_width5_proved.py` does. The resulting values of $T(N,5)$ for $N \le 40$ are recorded in `data/T5.txt`. They are complete, not merely stable under an increasing cutoff.

6 Symmetry at width five

The dihedral group D_5 acts on width-5 quiddity cycles by rotation and reversal.

Theorem 6.1. *The rotation action of* $\mathbb{Z}/5$ *on positive rational width-*5 *quiddity cycles is free. In particular* $5 \mid T(N,5)$ *for every* N.

Proof. The closing condition at width 5 is $(a_j a_{j+1} - 1)(a_{j+1} a_{j+2} - 1) = 1 + a_{j+1}$ for all j modulo 5. A cycle fixed by a nontrivial rotation is constant, say $a_j = a$, whence $(a^2-1)^2 = 1 + a$, that is $a(a^3 - 2a - 1) = 0$. Since $a > 0$ and $a^3 - 2a - 1 = (a+1)(a^2 - a - 1)$, the only positive root is the golden ratio $(1+\sqrt{5})/2$, which is irrational. So no positive rational cycle is fixed. □

7 Data and questions

Every value below is produced by a procedure whose search range is proved, so the counts are complete rather than stable under an increasing cutoff.

	$n=3$	$n=4$	$n=5$	$n=6$
$N=1$	1	2	5	14
$N=2$	1	4	20	102
$N=3$	1	6	40	259
$N=4$	1	6	60	487

Neither $T(N,5)$ nor $T(N,6)$ appears in the OEIS as a function of N, and neither is multiplicative: $T(2,5)T(3,5) = 800 \ne 110 = T(6,5)$.

At widths 8 and 9 the same reduction, applied with the glide symmetry as a pruning rule and swept over the entry box of Cuntz and Holm [3], gives

N	1	2	3	4
$T(N,8)$	132	2576	9980	28604
$T(N,9)$	429	13101		

(`code/w89_count.rs`). The values at $N = 1$ are C_6 and C_7, as they must be, and neither sequence appears in the OEIS. The sweep is complete over a proved box, so these are counts rather than stable observations; the cost grows like $N^{3(n-3)}$, which is why the table stops where it does. The reduction and the search bound behind it are routine and are carried out in the accompanying library rather than here.

Theorem 7.1 (Reduced form)**.** *For every $N \geq 1$,*

$$T(N,5) = \#\big\{(u,v,M) \in \mathbb{Z}^3_{>0} \;:\; \gcd(u,v)=1,\; M \mid N^2,\\ uv \mid N(u+v)+M,\; Nuv \mid M\big(N(u+v)+M\big)\big\}.$$

Proof. Multiplying $gu > N$ by $v > 0$ gives $guv > Nv$, that is $Nu + M > 0$, which holds for $u, M > 0$; the same argument disposes of $gv > N$. So both positivity conditions are automatic. For the equation, g is determined by (u,v,M), so requiring some g with $guv = N(u+v)+M$ is requiring $uv \mid N(u+v)+M$. Given that, multiplying $N \mid gM$ by uv gives $Nuv \mid M(N(u+v)+M)$, and the two are equivalent because $uv \neq 0$. $\square$

Proposition 7.2 (Hyperbola bound)**.** *Every solution counted by Theorem 7.1 satisfies $(u-N)(v-N) \leq N^2 + M \leq 2N^2$, and if $u \leq v$ then $u < 3N$.*

Proof. Since uv divides the positive integer $N(u+v)+M$ it is at most that integer, and $(u-N)(v-N) = uv - N(u+v) + N^2$, so $(u-N)(v-N) \leq N^2 + M$; and $M \leq N^2$ because $M \mid N^2$. If $u \leq N$ then $u < 3N$. Otherwise $0 < u - N \leq v - N$, so $(u-N)^2 \leq (u-N)(v-N) \leq 2N^2$, whereas $u \geq 3N$ would force $(u-N)^2 \geq 4N^2$. $\square$

The change is one of order: the smaller parameter is bounded *linearly* in N, where Lemma 5.2 gives only $N^3 + 2N^2$ for the original parameters. The bound $(u-N)(v-N) \leq 2N^2$ is attained for every $2 \leq N \leq 24$ (`code/t5_bounds.py`).

Corollary 7.3. $T(N,5) \;\leq\; 2 \sum_{M \mid N^2} \sum_{1 \leq u < 3N} d(Nu+M)$, *and consequently* $T(N,5) = O_\varepsilon\big(N^{1+\varepsilon}\big)$.

Proof. By Proposition 7.2 the smaller of u, v is less than $3N$; fix it as u and let v run over the divisors of $Nu+M$, which is legitimate because $v \mid N(u+v)+M$ and $v \mid Nv$ together give $v \mid Nu+M$. Each unordered pair is counted at most twice. For the second statement, $d(n) = O_\varepsilon(n^\varepsilon)$ gives $d(N^2) = O_\varepsilon(N^\varepsilon)$ and $d(Nu+M) = O_\varepsilon(N^\varepsilon)$ uniformly for $u < 3N$ and $M \leq N^2$, so the double sum is $O_\varepsilon(N^{1+\varepsilon})$. $\square$

No matching lower bound of the same order appears to exist.

Definition 7.4. Let p be prime. In the parametrisation of Theorem 7.1 the parameter M divides p^2, so $M \in \{1, p, p^2\}$, and $M = 1$ cannot occur: the condition $Nuv \mid M(N(u+v)+M)$ reads $p\,uv \mid p(u+v)+1$ there, forcing $p \mid 1$. Write $A(p)$ for the number of solutions with $M = p$ and $B(p)$ for the number with $M = p^2$, so that $T(p,5) = A(p) + B(p)$. At $M = p$ both divisibility conditions of Theorem 7.1 reduce to $uv \mid p(u+v+1)$, so

$$A(p) = \#\{(u,v) \in \mathbb{Z}^2_{>0} : \gcd(u,v) = 1,\; uv \mid p(u+v+1)\}.$$

A quiddity cycle has $M = p$ exactly when its parameters do, so this is also the count of cycles with $M = p$ used in Theorem 7.7.

Proposition 7.5. *Let $p, u, v \geq 1$ with $\gcd(uv, p) = 1$ and $uv \mid p(u + v + 1)$. Then*

$$(u, v) \in \{(1,1),\ (1,2),\ (2,1),\ (2,3),\ (3,2)\},$$

independently of p. These five pairs form a single orbit of the rotation action of $\mathbb{Z}/5$ of Theorem 6.1; that last assertion is not proved here and nothing below depends on it.

Proof. Since $u \mid uv \mid p(u + v + 1)$ and $\gcd(u, p) = 1$ we get $u \mid u + v + 1$, hence $u \mid v + 1$; symmetrically $v \mid u + 1$. So $u \leq v + 1$ and $v \leq u + 1$. If $u = v$ then $u \mid u + 1$ gives $u = 1$. If $v = u + 1$ then $u \mid v + 1 = u + 2$ gives $u \mid 2$, so (u, v) is $(1, 2)$ or $(2, 3)$; the case $u = v + 1$ is symmetric. Coprimality of u and v is not needed. □

So every further solution at a prime requires $p \mid uv$, and $T(p, 5)$ stays small: it is 160 at $p = 541$ and 130 at $p = 421$, against $T(120, 5) = 4170$.

By Remark 3.2 the width-5 glide gives $a_{j+3} = a_j a_{j+1} - 1$ for every j modulo 5, which in numerators is

$$p_j p_{j+1} = p\,(p_{j+3} + p) \qquad (j \in \mathbb{Z}/5), \tag{6}$$

so $e_j = p\,p_{j+3}$ and therefore $M_j = p\,p_{j+3}/g_j$ with $g_j = \gcd(p_j + p,\ p_{j+1} + p)$. In particular $M_j = p^2$ forces $p \mid p_{j+3}$, so the M-profile is controlled by which entries of the quiddity are integers.

Proposition 7.6. *Let p be prime. The number of integer entries in the quiddity cycle of a positive width-5 frieze over $\frac{1}{p}\mathbb{Z}$ is either 3 or 5, and it is 5 exactly for the Conway–Coxeter frieze $(2, 2, 1, 3, 1)$ and its rotations.*

Proof. Put $I = \{j : p \mid p_j\}$. If $p \nmid p_{j+3}$ then the right side of (6) has p-valuation exactly one, so exactly one of p_j, p_{j+1} is divisible by p, and that one has valuation exactly one. Taking $j = m + 2$, this says: for every $m \notin I$, exactly one of $m + 2$, $m + 3$ lies in I. Exhausting the 32 subsets of $\mathbb{Z}/5$, the only ones with that property have cardinality 3 or 5, and those of cardinality 3 are exactly the sets $\{i, i + 1, i + 3\}$. If $|I| = 5$ every entry is an integer, so the frieze is a Conway–Coxeter frieze of width 5, and there are $C_3 = 5$ of those, forming one rotation orbit. □

Theorem 7.7 (Orbit split)**.** *Let p be prime and write $R = T(p, 5)/5$ for the number of rotation orbits. Each orbit contains exactly two cycles with $M = p$ and three with $M = p^2$, except the orbit of the Conway–Coxeter frieze $(2, 2, 1, 3, 1)$, where all five have $M = p$. Hence $A(p) = 2R+3$ and $B(p) = 3R - 3$, so that*

$$3A(p) - 2B(p) = 15 \qquad \text{and} \qquad T(p, 5) = \tfrac{1}{2}\big(5A(p) - 15\big).$$

Proof. By Proposition 7.6, $I = \{j : p \mid p_j\}$ has three or five elements. If it has five the frieze is $(2, 2, 1, 3, 1)$ up to rotation, and evaluating M_j at each of the five positions gives p every time.

Suppose $|I| = 3$. The condition established in the proof of Proposition 7.6, that every $m \notin I$ has exactly one of $m+2$, $m+3$ in I, forces $I = \{i, i+1, i+3\}$ for some i. That proof also records that the divisible one of p_j, p_{j+1} has valuation exactly one whenever $p \nmid p_{j+3}$; applying this at $j = i-1$ and at $j = i+1$, where p_{i+2} respectively p_{i+4} is prime to p, gives $v_p(p_i) = v_p(p_{i+1}) = 1$ exactly, while $v_p(p_{i+2}) = v_p(p_{i+4}) = 0$ because $i + 2, i + 4 \notin I$. Now take the five positions in turn.

For $j = i+1$ and $j = i+4$ we have $j+3 \notin I$. In each case one of the two entries p_j, p_{j+1} is prime to p, hence so is g_j, and $v_p(M_j) = 1+0-0 = 1$, so $M_j = p$.

For $j = i+2$ and $j = i+3$ the entry p_{i+2} respectively p_{i+4} is prime to p, so again g_j is prime to p, while $v_p(p_{j+3}) = 1$; thus $v_p(M_j) = 2$ and $M_j = p^2$.

For $j = i$ write $p_i = p\alpha$ and $p_{i+1} = p\beta$, so $g_i = pG$ with $G = \gcd(\alpha+1, \beta+1)$, and (6) at $j = i$ gives $p_{i+3} = p(\alpha\beta - 1)$ with $\alpha\beta - 1 > 0$. If $M_i = p$ then $g_i = p_{i+3}$, that is $G = \alpha\beta - 1$. Since $G \mid \alpha + 1$ this makes $\alpha\beta - 1$ divide $\alpha + 1$, say $\alpha + 1 = (\alpha\beta - 1)s$. But (6) at $j = i+2$ reads $p_{i+2}p_{i+3} = p(p_i + p)$, that is $p_{i+2}(\alpha\beta - 1) = p(\alpha + 1)$, so $p_{i+2}(\alpha\beta - 1) = p(\alpha\beta - 1)s$, and cancelling $\alpha\beta - 1 \neq 0$ gives $p_{i+2} = ps$, contradicting $p \nmid p_{i+2}$. Hence $M_i = p^2$.

So two positions carry $M = p$ and three carry $M = p^2$. Counting over the R orbits, $A(p) = 2(R-1) + 5$ and $B(p) = 3(R-1)$, and $T(p, 5) = A + B = 5R$. □

8 The count at every N

Theorem 7.1 counts triples (u, v, M) with M running over the divisors of N^2. The parameter can be removed from each term at the cost of a sum: what remains at each contributing M is an instance of the equation of Ford's problem.

Definition 8.1. For $A, B \geq 1$ put

$$F(A, B) = \#\{(u, v) \in \mathbb{Z}^2_{>0} : \gcd(u, v) = 1,\ uv \mid A(u+v) + B\},$$

the number of coprime solutions of $huv = A(u+v) + B$, the equation of Ford's problem with those parameters.

Theorem 8.2. *Let $N \geq 1$. For $M \mid N^2$ write $e = \gcd(N, M)$ and $d = N/e$. Then*

$$T(N, 5) = \sum_{\substack{M \mid N^2 \\ N \mid M^2}} F\Big(e,\ \frac{M}{d}\Big),$$

and the divisors omitted from the sum contribute nothing. The condition $N \mid M^2$ is the condition $d \mid M$: writing $M = eM'$, both say $d \mid e$, since $\gcd(d, M') = 1$.

Proof. Fix $M \mid N^2$ and a coprime pair (u, v). The first condition of Theorem 7.1 says that $g := \big(N(u+v) + M\big)/uv$ is an integer, and given that, the second is $N \mid gM$. Write $M = eM'$; then $\gcd(d, M') = 1$, so $N \mid gM$ is $ed \mid geM'$, that is $d \mid gM'$, that is $d \mid g$.

Put $g = dh$. Then $dhuv = ed(u+v) + M$, and since $d \mid ed(u+v)$ this forces $d \mid M$, which is the support condition; dividing by d leaves

$$huv = e(u+v) + \frac{M}{d}. \tag{7}$$

Conversely, if $\gcd(u, v) = 1$ and (7) holds for some $h \geq 1$, then $g := dh$ satisfies $guv = N(u+v) + M$ and $d \mid g$, hence $N \mid gM$. The positivity conditions are automatic, by the argument of Theorem 7.1. So for each M in the support the pairs counted by Theorem 7.1 correspond bijectively to those counted by $F(e, M/d)$, and the sum over M gives the identity. □

The index set has a second description, as a set of friezes.

Theorem 8.3. *Let $N \geq 1$. A pair (x, x) satisfies* (5) *if and only if $x > N$, $N \mid x^2$ and $x - N \mid N^2$. The map $M \mapsto M + N$ is a bijection from the index set of Theorem 8.2 onto these pairs. Hence the sum has $Q(N)$ terms, where*

$$Q(N) = \prod_p \big(\lfloor 3a_p/2 \rfloor + 1\big), \qquad N = \prod_p p^{a_p},$$

the number of square divisors of N^3 (OEIS A092520).

Proof. Positivity of the pair is $x > 0$, and $N^2 < x^2$ is $x > N$. Writing $x = N + k$ with $k \geq 1$, $x^2 - N^2 = k(k+2N)$ and $x + N = k + 2N$, so the two divisibility conditions of (5) coincide and read $k \mid N^2$; and $N \mid x^2$ is the remaining one. Substituting $x = M + N$, the conditions become $M \mid N^2$ and, since $(M+N)^2 = M^2 + N(2M+N)$, also $N \mid M^2$. These are the conditions of Theorem 8.2, so the map is a bijection.

For the count, take $N = p^a$ and $M = p^m$: then $M \mid N^2$ reads $m \leq 2a$ and $N \mid M^2$ reads $\lceil a/2 \rceil \leq m$, leaving $2a - \lceil a/2 \rceil + 1 = \lfloor 3a/2 \rfloor + 1$ values. Both conditions are multiplicative in N, and so is the count. $\square$

The pairs of Theorem 8.3 are exactly the quiddity cycles fixed by the reflection of the dihedral action, so the decomposition carries one Ford term for each reflection-symmetric frieze.

At a prime the sum has two terms. The divisors of p^2 are $1, p, p^2$; for $M = 1$ one has $d = p$, which does not divide 1, while $M = p$ and $M = p^2$ both give $e = p$ and $d = 1$. So

$$T(p, 5) = F(p, p) + F(p, p^2) = A(p) + B(p),$$

the split of Definition 7.4, and Theorem 7.7 evaluates it as $5 + 5C(p)$. That collapse is special to primes. For composite N the sum has more terms and no single instance of Ford's equation carries the count; in particular the guess $T(N, 5) = 5 + 5C(N)$, with $C(N)$ the solution count of $auv = N + u + v$, is false, already at $N = 12$, where the two sides are 310 and 40.

Theorem 8.4. *$T(N, 5) = O_\varepsilon\big(N^{2/3+\varepsilon}\big)$.*

Proof. Fix a term of Theorem 8.2, with parameters $A = e \leq N$ and $B = M/d \leq N^2$, and let (h, u, v) be counted by $F(A, B)$, so $huv = A(u + v) + B$. Put $T = \min(h, u, v)$. Taking $u \leq v$, which is no loss since the equation is symmetric in u and v, one has $huv \leq 2Av + B$, while $T^2 \leq hu$ and $T \leq v$, so $T^2 v \leq huv \leq 2Av + B$ and hence $v(T^2 - 2A) \leq B$. If $T^2 \leq 2A$ then $T^3 \leq 2AT$; otherwise $T \leq v$ gives $T(T^2 - 2A) \leq B$. Either way

$$T^3 \leq 2AT + B. \tag{8}$$

With $A \leq N$ and $B \leq N^2$ this forces $T^3 \leq 8N^2$. Indeed if $2T \leq N$ then $2NT \leq N^2$ and $T^3 \leq 2N^2$; and if $2T > N$ then $N^2 < 2NT$, so $T^3 < 4NT$, whence $T^2 < 4N$ and $T^6 < 64N^3 \leq 64N^4$.

Now sort the solutions by which of h, u, v is least. If u is least then $v(hu - A) = Au + B$, so v divides $Au + B$; the case of v is symmetric. If h is least then $(hu - A)(hv - A) = A^2 + Bh$, so $hu - A$ divides $A^2 + Bh$. In each case the least variable is at most $2N^{2/3}$ by the previous paragraph, and it determines the remaining two up to the number of divisors of an integer that is at most a fixed power of N. Since $d(n) = O_\varepsilon(n^\varepsilon)$,

$$F(A, B) = O_\varepsilon\big(N^{2/3+\varepsilon}\big).$$

By Theorem 8.3 there are $N^{o(1)}$ terms, which gives the theorem. $\square$

The least variable does reach order $N^{2/3}$, so no sharper exponent follows from (8) alone. Theorem 8.4 improves Corollary 7.3, which gives $O_\varepsilon(N^{1+\varepsilon})$, and it uses only the divisor bound, not the Shiu-type input of Theorem 9.3. At a prime Theorem 9.3 remains sharper, at $O_\varepsilon(p^{1/3+\varepsilon})$, because the substitution of Lemma 9.1 lowers the parameter B from p^2 to p there.

Theorem 8.5. *A pair (A, B) of positive integers occurs as the parameters of a term of Theorem 8.2, for some N, if and only if*

$$B \mid \gcd(A, B)^2.$$

When it does, it occurs at $N = d^2 f$ with $f = \gcd(A, B)$ and $d = A/f$, so at some $N \le A^2$.

Proof. Suppose (A, B) occurs, with $A = \gcd(N, M)$, $d = N/A$ and $B = M/d$. The support condition gives $d \mid M$, and $d \mid N$, so $d \mid \gcd(N, M) = A$; write $A = df$. Then $N = d^2 f$, and with $M' = M/A$ one has $\gcd(d, M') = 1$ and $B = fM'$, so $\gcd(A, B) = f \gcd(d, M') = f$. From $M \mid N^2$, that is $dfM' \mid d^4 f^2$, cancelling df gives $M' \mid d^3 f$, and $\gcd(M', d) = 1$ gives $M' \mid f$. Hence $B = fM' \mid f^2 = \gcd(A, B)^2$.

Conversely, suppose $B \mid \gcd(A, B)^2$ and put $f = \gcd(A, B)$, $d = A/f$, $m = B/f$. Then $\gcd(d, m) = 1$ and $m \mid f$. Take $N = d^2 f$ and $M = dfm$. Then $\gcd(N, M) = df \gcd(d, m) = A$, so $N/\gcd(N, M) = d$, which divides M; and $M \mid N^2$ because $m \mid f$. Finally $M/d = fm = B$. Since $f \mid A$ we have $N = d^2 f = A^2/f \le A^2$. □

Theorem 8.5 turns the relation with Ford's problem from an implication into an equivalence.

Corollary 8.6. *Conjecture 9.5 holds if and only if $F(A, B) = A^{o(1)}$ for every pair with $B \mid \gcd(A, B)^2$.*

Proof. If the bound on F holds then, by Theorem 8.2, $T(N, 5)$ is a sum of $N^{o(1)}$ terms by Theorem 8.3, each $F(A, B)$ with $A \le N$ and, by Theorem 8.5, with $B \mid \gcd(A, B)^2$; so $T(N, 5) = N^{o(1)}$. Conversely, given such a pair, Theorem 8.5 places $F(A, B)$ as one term of $T(N, 5)$ for some $N \le A^2$, and every term is at most the whole sum, so $F(A, B) \le T(N, 5) = N^{o(1)} = A^{o(1)}$. □

So the width-five count is not merely an instance of Ford's problem: it is a faithful reformulation of the subfamily $B \mid \gcd(A, B)^2$, and settling one settles the other. The subfamily is proper, since $\gcd(A, B)^2$ can be smaller than B; the case $A = B = p$, which is $A(p)$, lies in it, and so does $A = p$, $B = p^2$, which is $B(p)$.

Theorems 8.2 and 8.5 carry information from the Diophantine side to the frieze side. The transfer also runs the other way, and gives a statement about Ford's equation that its shape does not suggest.

Theorem 8.7. *For every $N \ge 1$,*

$$T(N, 5) \equiv 5\, Q(N) \pmod{10},$$

with Q as in Theorem 8.3. Hence $10 \mid T(N, 5)$ if and only if $Q(N)$ is even, that is, if and only if some exponent in the factorisation of N is congruent to 1 or 2 modulo 4.

Proof. Reversal of the quiddity cycle acts on the pairs of (5) by $(p, q) \mapsto (q, p)$, an involution whose fixed points are the pairs (x, x). So $T(N, 5)$ and the number of those pairs have the same parity, and by Theorem 8.3 that number is $Q(N)$. With $5 \mid T(N, 5)$ from Theorem 6.1, and $5Q(N) \equiv Q(N) \bmod 2$, the congruence follows. For the last clause, $\lfloor 3a/2 \rfloor + 1$ is even exactly when $a \equiv 1, 2 \pmod 4$, so $Q(N)$ is odd exactly when every exponent is $\equiv 0, 3 \pmod 4$. □

At $N = p$ this gives $10 \mid T(p,5)$, since $a = 1$. The first N with $10 \nmid T(N,5)$ and $N > 1$ is $N = 8$, where $Q(8) = 5$ and $T(8,5) = 145$.

Corollary 8.8. *For every $N \geq 1$,*

$$5 \;\Bigg|\; \sum_{\substack{M \mid N^2 \\ d \mid M}} F\Big(\gcd(N,M),\ \frac{M}{d}\Big), \qquad d = \frac{N}{\gcd(N,M)}.$$

For squarefree N this reads $5 \mid \sum_{m \mid N} F(N, Nm)$.

Proof. The sum is $T(N,5)$ by Theorem 8.2, and $5 \mid T(N,5)$ by Theorem 6.1. $\square$

The summands are not themselves divisible by 5. At $N = 8$ the five terms are 17, 27, 31, 31, 39, and they sum to 145; over $N \leq 25$ only 23 of the 101 terms are divisible by 5, while all 25 sums are (`code/ford_congruence.py`). Each term is odd, since $(u,v) \mapsto (v,u)$ fixes only $u = v = 1$, so by Theorem 8.3 the sum of $Q(N)$ odd numbers is congruent to $Q(N)$ modulo 2, which is Theorem 8.7 again.

9 The width-five count at a prime

No closed form for $T(p,5)$ should be expected. Setting $u = ps$ in the $M = p$ term and writing $v + 1 = st$, the condition $uv \mid p(u+v+1)$ becomes $st - 1 \mid p + t$, and putting $p + t = (st-1)m$ turns it into the symmetric cubic

$$stm - t - m = p, \qquad \text{equivalently} \qquad (st-1)(sm-1) = sp + 1. \tag{9}$$

The correspondence is exact.

Lemma 9.1. *Let $p \geq 5$ be prime and let $C(p)$ be the number of triples (s,t,m) of positive integers with $stm = p + t + m$. Then $A(p) = 5 + 2C(p)$.*

Proof. By Definition 7.4, $A(p)$ counts the coprime pairs (u,v) with $uv \mid p(u+v+1)$, a condition symmetric in u and v. If $\gcd(uv,p) = 1$ it reduces to $uv \mid u+v+1$, and Proposition 7.5 lists the five pairs satisfying it. Otherwise $p \mid uv$, and $\gcd(u,v) = 1$ forces p to divide exactly one of u and v; by the symmetry the two halves are equinumerous, so $A(p) - 5 = 2\,\#\{(u,v) : p \mid u\}$.

For the pairs with $p \mid u$ write $u = ps$. Then $uv \mid p(u+v+1)$ reads $sv \mid ps + v + 1$, whence $s \mid v+1$; put $v = st-1$ with $t \geq 1$. The condition becomes $s(st-1) \mid s(p+t)$, that is $st-1 \mid p+t$; put $p + t = (st-1)m$ with $m \geq 1$. Expanding, $stm = p + t + m$, which is (9).

Conversely, given (s,t,m) with $stm = p + t + m$, set $u = ps$ and $v = st - 1$. Then $st - 1 = (p+t)/m \geq 1$, so $v \geq 1$, and $\gcd(u,v) = 1$: indeed $\gcd(s, st-1) = 1$, and $p \nmid st - 1$, since $st \equiv 1 \bmod p$ would give $m \equiv t + m$, hence $p \mid t$ and then $st \equiv 0$, a contradiction. The two maps are mutually inverse, so $\#\{(u,v) : p \mid u\} = C(p)$. $\square$

Checked against $A(p)$ computed directly for the primes $5 \leq p \leq 53$ (`code/a_count.py`), with no discrepancy. Equation (9) also bounds the count, and does so better than Corollary 7.3.

The exponent can be lowered to $\frac{1}{3}$ by importing the method of Conrey and Shah [7], who bound the number of representations $n = xyz + x + y + z$ by $n^{1/3} \log n (\log\log n)^4$. Their argument uses that full symmetry forces the smallest variable below $n^{1/3}$. Our equation $suv = p + u + v$ is symmetric only in u and v, but the same bound holds.

Lemma 9.2. *For $p \geq 4$, every solution of $suv = p+u+v$ in positive integers has* $\min(s,u,v)^3 \leq 2p$.

Proof. If $\min(s,u,v)^3 > 2p$ then $suv > 2p$, so $p < u+v$ and hence $suv = p+u+v < 2(u+v) \leq 4\max(u,v)$. Cancelling the maximum leaves a product of two of the variables below 4. But $\min(s,u,v)^3 > 2p \geq 8$ forces every variable to be at least 3, so that product is at least 9. □

Theorem 9.3. $T(p,5) = O_\varepsilon\big(p^{1/3+\varepsilon}\big)$.

Proof. By Lemma 9.2 some variable is at most $(2p)^{1/3}$. If it is $s = t$, then $(tu-1)(tv-1) = tp+1$ determines (u,v) from a divisor of $tp+1$. If it is $u = t$ or $v = t$, then $v(su-1) = p+u$ determines the pair from a divisor of $p+t$. Hence

$$C(p) \;\leq \sum_{1\leq t\leq (2p)^{1/3}} \big(d(tp+1) + 2\,d(p+t)\big),$$

and a divisor-sum bound of Shiu type [8] gives $\ll p^{1/3}\log p\,(\log\log p)^c$. The conclusion follows from $T(p,5) = 5 + 5C(p)$ of Theorem 7.7. □

For all 123 primes $5 \leq p < 700$ the ratio $T(p,5)/\sqrt{p}$ has mean 15.2, and 12.0 over $p > 400$ (`code/t5_primes_split.py`); it does not decay on that range. Theorem 9.3 forces it to decay eventually, and the paragraph above places the onset near $p = 10^7$, so the small-prime data says nothing against it.

The conjecture is not new, and it is not ours. Conrey asked whether the number of positive integer solutions of $xyz + x + y = n$ is $O_\varepsilon(n^\varepsilon)$, and Ford generalised this to $xyz = A(x+y) + B$ with $O_\varepsilon(|AB|^\varepsilon)$ solutions; both questions are recorded in [9]. Our width-5 equation $guv = N(u+v) + M$ is Ford's equation with $A = N$ and $B = M$, so Conjecture 9.5 below is a case of Ford's problem, summed over $M \mid N^2$. Writing $R_a(n) = \#\{(x,y) \in \mathbb{N}^2 : axy - x - y = n\}$ for the quantity studied in [9], at a prime

$$C(p) = \sum_{a\geq 1} R_a(p), \qquad R_1(p) = d(p+1),$$

both verified for every prime $p < 200$. The second identity is Proposition 9.6: our lower bound is the $a = 1$ term. Huang proves an asymptotic for $\sum_{n\leq N} R_a(n)$ with a fixed, which is the averaged form of the inner sum.

That the frieze count is governed by a problem of Conrey and Ford is the reason Conjecture 9.5 is stated and not proved here.

9.1 The count as a divisor sum in progressions

The cubic factors on each slice, which locates the analytic difficulty precisely.

Proposition 9.4. *Let $n \geq 1$ and $a \geq 1$. The solutions of $auv = n + u + v$ with that value of a correspond bijectively to the divisors d of $an + 1$ with $d \equiv -1 \pmod{a}$, by $d = au - 1$. Consequently*

$$C(n) = \sum_{a=1}^{n+2} \#\{d \mid an+1 \;:\; d \equiv -1 \pmod{a}\}.$$

The slice factorisation is the standard one for this equation and underlies the mean value theorem of [9]; it is recorded here in the form used below and because it is what the formalization verifies.

Proof. Multiplying $auv = n + u + v$ by a gives $(au - 1)(av - 1) = a(auv - u - v) + 1 = an + 1$, and conversely dividing by $a \neq 0$ recovers the equation, so the two are equivalent. If (u, v) is a solution then $d = au - 1$ divides $an + 1$ and $d \equiv -1 \pmod{a}$. Conversely let $d \mid an + 1$ with $d \equiv -1 \pmod{a}$ and put $e = (an+1)/d$. Then $de \equiv 1 \pmod{a}$ and $d \equiv -1$, so $e \equiv -1 \pmod{a}$; hence $u = (d+1)/a$ and $v = (e+1)/a$ are positive integers and $(au-1)(av-1) = de = an+1$. The two constructions are mutually inverse. Finally $u, v \geq 1$ forces $a \leq n + 2$, attained at $u = v = 1$. $\square$

So counting $C(n)$ is counting divisors of $an + 1$ in one residue class modulo a, summed over a. The modulus and the quantity move together: at a of size $n^{1/2}$ the modulus a is comparable to the square root of $an + 1$, which is exactly the range in which the equidistribution results for divisors in progressions give nothing. Proposition 9.8 records the same obstruction in its modular-hyperbola form, where the Weil error is of size $p^{1/2}$ against a quantity of size $p^{1/3}$. The analytic half of the problem is therefore not a missing trick: it is the divisor-in-progressions problem at the modulus where the known technology stops.

Conjecture 9.5. $T(N, 5) = N^{o(1)}$.

The natural sharper guess, $T(N, 5) \asymp d(N^2)\log^2 N$, is false, and the reason is a lower bound that the numerics cannot see.

Proposition 9.6. *$C(p) \geq d(p+1)$, hence $T(p, 5) \geq 5\,d(p+1)$, for every prime p. Consequently $T(N, 5) \asymp d(N^2)\log^2 N$ fails.*

Proof. Taking $u = 1$ in (9), written as $suv = p + u + v$, leaves $sv = p + 1 + v$, which is solvable exactly when $v \mid p + 1$, with $s = (p + 1)/v + 1$. Distinct divisors give distinct solutions, so $C(p) \geq d(p + 1)$, and $T(p, 5) = 5 + 5C(p)$ by Theorem 7.7.

For the consequence, let m_r be the product of the first r primes, so $d(m_r) = 2^r$ and $\log m_r \sim r \log r$. By Linnik's theorem there is a prime $p \equiv -1 \pmod{m_r}$ with $\log p \ll \log m_r$. Then $m_r \mid p + 1$, so $d(p + 1) \geq 2^r$ while $\log^2 p \ll (r \log r)^2$, and $2^r/(r \log r)^2 \to \infty$. Since $d(p^2) = 3$ at a prime, the asserted equivalence would force $d(p + 1) = O(\log^2 p)$. $\square$

The crossover is far out of computational reach: 2^r overtakes $(r \log r)^2$ near $r = 20$, where m_r already exceeds 10^{25}. So the flat ratios observed for $N \leq 3000$ are what the proposition predicts, not evidence against it. This is why the equivalence are what the proposition predicts, not evidence against it: numerical stability over three decades is not an asymptotic.

Multiplying $auv = p + u + v$ through by a gives

$$(au - 1)(av - 1) = ap + 1, \tag{10}$$

so the solutions with leading coefficient a are the divisors of the shifted number $ap + 1$ lying in the class $-1 \bmod a$, and

$$C(p) = \sum_{a \geq 1} \#\{D \mid ap + 1 \,:\, D \equiv -1 \bmod a\}. \tag{11}$$

The term $a = 1$ is $d(p+1)$, the same count as the $u = 1$ family of Proposition 9.6 though a different family. The term $a = 2$ is the one that costs nothing: $2p+1$ is odd, so every divisor of it is odd and hence of the form $2u-1$, and no divisor is lost to the congruence. Thus $R_2(p) = d(2p+1)$ exactly, and since the two families carry different values of a they are disjoint.

Proposition 9.7. *$C(p) \geq d(p+1) + d(2p+1)$.*

Proof. Formula (10) at $a = 1$ and $a = 2$; the two families are disjoint because a is determined by the solution. Formalised as `cubic_lower_two_families`. □

9.2 The count on the modular hyperbola

The decomposition (11) can be run in the other direction, eliminating a instead of u and v. Setting $d = au - 1$ and $e = av - 1$, identity (10) reads $de = ap + 1$, so $de \equiv 1 \bmod p$ and $a = (de-1)/p$, while $a \mid d+1$ and $a \mid e+1$ are exactly the conditions defining u and v.

Proposition 9.8. *$C(p) = \#\big\{(d,e) \in \mathbb{Z}^2_{>0} : de \equiv 1 \bmod p,\ \frac{de-1}{p} \mid \gcd(d+1, e+1)\big\}$.*

Proof. The maps $(a,u,v) \mapsto (au-1, av-1)$ and $(d,e) \mapsto \big((de-1)/p, (d+1)p/(de-1), (e+1)p/(de-1)\big)$ are mutually inverse by (10). That the second is defined on the whole target set, which is where the divisibilities $(de-1)/p \mid d+1$ and $(de-1)/p \mid e+1$ are used, is checked rather than argued below, so this proposition rests on a computation at that point. Checked against $C(p)$ for all primes $5 \leq p < 120$ with no discrepancy (`code/modular_hyperbola.py`). □

So $C(p)$ counts points of the modular hyperbola $de \equiv 1 \bmod p$ cut by a divisibility that ties the level $a = (de-1)/p$ to $\gcd(d+1, e+1)$. The same substitution turns the sum that every upper bound passes through into a hyperbola count with no side condition,

$$\sum_{a \leq A} d(ap+1) = \#\{(d,e) \in \mathbb{Z}^2_{>0} : de \equiv 1 \bmod p,\ 1 < de \leq Ap+1\}, \tag{12}$$

and this locates the wall. Counts of points of $de \equiv 1 \bmod p$ in regions are governed by Kloosterman sums, and Weil's bound gives them with an error $O_\varepsilon(p^{1/2+\varepsilon})$. With $A = (2p)^{1/3}$ the main term of (12) is of size $A \log p$, that is $p^{1/3+o(1)}$. The error term exceeds the main term. Kloosterman technology therefore does not reach the trivial bound $C(p) \ll_\varepsilon p^{1/3+\varepsilon}$ of Theorem 9.3, let alone improve on it, and the improvement would have to come from below the square root barrier rather than from a sharper application of Weil.

The substitution (10) is an identity over any commutative ring (`fixed_point_ring`), which is why the analogue over $\mathbb{F}_q[t]$ inherits the structure of the problem intact rather than simplifying it.

10 The width-six reduced form

At width five the conditions reduce to Ford's equation in three variables (Theorem 7.1). The same reduction is available at width six, with one further variable, and it is again of Ford type.

Write the quiddity numerators of a width-six frieze over $\frac{1}{N}\mathbb{Z}$ as p, q, r for a_0, a_1, a_2, and f for the middle numerator Na_4. By (4) these satisfy

$$pqr = N^2(f + p + r), \qquad f \mid pq, \qquad f \mid qr, \qquad f \mid N(p+r). \tag{13}$$

Put

$$A = pq - N^2, \qquad B = qr - N^2, \qquad g = \gcd(A, B), \qquad U = A/g, \qquad V = B/g, \qquad R = q.$$

Theorem 10.1 (Master identity)**.** *$AB = N^4 + N^2 qf$, that is $g^2UV = N^4 + N^2Rf$.*

Proof. $AB = (pq-N^2)(qr-N^2) = q\,(pqr) - N^2(pq+qr) + N^4$. Substituting $pqr = N^2(f+p+r)$ gives $qN^2(f+p+r) - N^2(pq+qr) + N^4 = N^2qf + N^4$. $\square$

The variable R is not free. Since $pq = A + N^2$ and $qr = B + N^2$, the numerator q divides both $gU + N^2$ and $gV + N^2$, so R ranges over the divisors of $\gcd(gU + N^2, gV + N^2)$.

Theorem 10.2 (Reduced form at width six)**.** *The assignment $(p, q, r) \mapsto (g, U, V, R)$ is a bijection from the width-six friezes over $\frac{1}{N}\mathbb{Z}$ onto the quadruples of positive integers satisfying*

$$\gcd(U, V) = 1, \qquad N \mid gU, \qquad N \mid gV, \qquad R \mid gU + N^2, \qquad R \mid gV + N^2,$$
$$N^2R \mid g^2UV - N^4,$$

and, writing $f = (g^2UV - N^4)/(N^2R)$,

$$f \mid gU + N^2, \qquad f \mid gV + N^2, \qquad fR \mid N\big(g(U + V) + 2N^2\big).$$

The inverse is $p = (gU + N^2)/R$, $q = R$, $r = (gV + N^2)/R$.

Proof. Let a width-six frieze be given. The row below the quiddity has entries $a_ja_{j+1} - 1$; at $j = 0$ this is A/N^2, which lies in $\frac{1}{N}\mathbb{Z}$ exactly when $N \mid A$, and at $j = 1$ it is B/N^2, giving $N \mid B$. These are the conditions $N \mid gU$ and $N \mid gV$. Integrality of p and r gives $R \mid gU + N^2$ and $R \mid gV + N^2$, and Theorem 10.1 gives $N^2R \mid g^2UV - N^4$ with quotient f. The three divisibilities of (13) are the last three conditions, since $pq = gU + N^2$, $qr = gV + N^2$ and $N(p + r)R = N(g(U + V) + 2N^2)$. As $\gcd(A, B) = g$ we have $\gcd(U, V) = 1$.

Conversely, let a quadruple satisfy the conditions and define p, q, r, f by the stated formulas. Then $pR = gU + N^2$ and $rR = gV + N^2$, so

$$(pR)(rR) = g^2UV + N^2g(U + V) + N^4 = (N^4 + N^2Rf) + N^2g(U + V) + N^4,$$

while $g(U + V) + 2N^2 = (p + r)R$ gives $N^2g(U + V) = N^2(p + r)R - 2N^4$. Substituting and dividing by R yields $prR = N^2(f + p + r)$, which is $pqr = N^2(f + p + r)$. The remaining conditions of (13) are the last three hypotheses, and $\gcd(A, B) = g\gcd(U, V) = g$, so the two assignments are mutually inverse. $\square$

The conditions $N \mid gU$ and $N \mid gV$ are exactly the lattice condition on the row fixed by the glide, as in Remark 3.2; without them (13) admits solutions that are not friezes, for example $(p, q, r, f) = (1, 5, 8, 1)$ at $N = 2$, where $A = pq - N^2 = 1$ is not divisible by N. The system is checked against the enumeration for $N \le 8$ by `code/width6_reduced.py`.

11 Friezes over other coefficient rings

Nothing in the width-five reduction uses an order. Over any field the parameterisation (3) closes after five steps, and over $\frac{1}{N}\mathcal{O}$ for a commutative domain $\mathcal{O}$ the conditions of Proposition 5.1 are the same divisibilities read in $\mathcal{O}$. Positivity selects solutions; it is not part of the reduction.

Two conditions that positivity hides must then be imposed separately.

Proposition 11.1. *In* (3) *the entry* a_2 *vanishes exactly when* $a_0 = -1$*, and* a_4 *vanishes exactly when* $a_1 = -1$*.*

Proof. $a_2 = (a_0 + 1)/D$ and $a_4 = (a_1 + 1)/D$ with $D \neq 0$. □

Over an ordered field with all entries positive these cases cannot occur, which is why the real theory does not record them. Over a general domain they are genuine conditions: at $p = -N$ the requirement $(pq - N^2) \mid N^2(p + N)$ is vacuous, the solution set is infinite, and the array has a zero entry.

Theorem 11.2 (Ford's equation over a domain)**.** *Let* $\mathcal{O}$ *be a commutative domain and let* $N, p, q, k, l \in \mathcal{O}$ *satisfy* $D := pq - N^2 \neq 0$*,* $N^2(p + N) = Dk$ *and* $N^2(q + N) = Dl$*. Then*

$$Dkl = N^4 + N^3(k + l).$$

Proof. Multiplying the two hypotheses gives $D^2kl = N^4(p + N)(q + N)$, and adding them gives $N^2(p + q) + 2N^3 = D(k + l)$. Since $(p + N)(q + N) = D + 2N^2 + N(p + q)$,

$$D^2kl = N^4\bigl(D + 2N^2 + N(p + q)\bigr) = N^4D + 2N^6 + N^3\bigl(D(k + l) - 2N^3\bigr) = D\bigl(N^4 + N^3(k + l)\bigr),$$

and cancelling D gives the identity. □

Theorem 11.3 (Finiteness over the Gaussian integers)**.** *Let* $\|\cdot\|$ *be the norm on* $\mathbb{Z}[i]$*. A solution of* $Dkl = N^4 + N^3(k + l)$ *over* $\mathbb{Z}[i]$ *with* $D \neq 0$ *satisfies*

$$\min\bigl(\|k\|, \|l\|\bigr) \leq 8N^6 + 2N^8.$$

Moreover $l\,(Dk - N^3) = N^4 + N^3k$*, so* l *is determined by* D *and* k *unless* $Dk = N^3$*, in which case* $k = -N$ *and* $p = 0$*. The set of width-five friezes over* $\frac{1}{N}\mathbb{Z}[i]$ *is therefore finite.*

Proof. The norm is multiplicative and satisfies $\|x + y\| \leq 2\|x\| + 2\|y\|$, since $(x_1 + y_1)^2 + (x_2 + y_2)^2 \leq 2(x_1^2 + x_2^2) + 2(y_1^2 + y_2^2)$. As $D \neq 0$ we have $\|D\| \geq 1$, so with $\alpha = \|k\|$ and $\beta = \|l\|$,

$$\alpha\beta \leq \|D\|\alpha\beta = \|N^4 + N^3(k + l)\| \leq 2N^8 + 2N^6\|k + l\| \leq 2N^8 + 4N^6(\alpha + \beta).$$

Suppose both α and β exceed $8N^6 + 2N^8$. Then $\alpha - 4N^6$ and $\beta - 4N^6$ both exceed $4N^6 + 2N^8$, so $\alpha\beta - 4N^6(\alpha + \beta) = (\alpha - 4N^6)(\beta - 4N^6) - 16N^{12} > 16N^{14} \geq 2N^8$, contradicting the display. The rearrangement $l(Dk - N^3) = N^4 + N^3k$ is immediate from the equation; if $Dk = N^3$ then $N^4 + N^3k = 0$, so $k = -N$, and then $N^2(p+N) = Dk = N^3$ gives $p = 0$, which is not a quiddity numerator, so that branch is empty. By Proposition 11.1 neither k nor l vanishes. The equation is symmetric in k and l, so assume $\|k\| \leq 8N^6 + 2N^8$. If $\|D\|\|k\| > 4N^6$ then the display gives $\|l\|\bigl(\|D\|\|k\| - 4N^6\bigr) \leq 2N^8 + 4N^6\|k\|$, so $\|l\|$ is bounded, and $D = (N^4 + N^3(k + l))/kl$ is then determined by k and l. Otherwise $\|D\| \leq 4N^6$ and $\|k\| \leq 4N^6$, and l is determined by D and k. Either way the solutions are finite in number. □

The counts of width-five friezes over $\frac{1}{N}\mathbb{Z}[i]$ are 55, 580 and 815 for $N = 1, 2, 3$ (`code/gaussian_width5.py`); the value at $N = 1$ is to be compared with $T(1, 5) = 5$ over $\mathbb{Z}$.

12 The Farey-path dictionary

The friezes counted here are classified in [5]. For positive integers N, K, R with $N = KR$, Theorem B of that paper puts the minimal clockwise paths of length n in the Farey graph $\mathcal{F}_R$, modulo $SL_2(\mathbb{Z})$, in one-to-one correspondence with the positive friezes over $\frac{1}{N}\mathbb{Z}$ of width n and greatest common divisor K, through $m_{i,j} = \frac{1}{R}(a_j b_i - b_j a_i)$. Here $\mathcal{F}_R$ has as vertices the integer pairs (a, b) with $\gcd(a, b) \mid R$ and a directed edge from (a, b) to (c, d) when $ad - bc = R$, and a path is minimal when the integers $a_j b_i - b_j a_i$ have greatest common divisor 1.

This section makes the passage between the two sides explicit, and does so at every width at once. Write $[u, w] = u_1 w_2 - u_2 w_1$ for $u, w \in \mathbb{Z}^2$.

Definition 12.1. The path attached to integers $c_0, c_1, \ldots$ and a pair $v_0, v_1 \in \mathbb{Z}^2$ is the sequence $v_{i+1} = c_i v_i - v_{i-1}$.

Lemma 12.2. *$[v_i, v_{i+1}] = [v_0, v_1]$ for every i.*

Proof. $[v_i, v_{i+1}] = [v_i, c_i v_i - v_{i-1}] = -[v_i, v_{i-1}] = [v_{i-1}, v_i]$, and induction on i. □

So one edge decides the graph: if $[v_0, v_1] = R$ then every edge of the path is an edge of $\mathcal{F}_R$, and no further checking is required.

Theorem 12.3. *Let $v_i \in \mathbb{Z}^2$ satisfy $[v_i, v_{i+1}] = R$ for all i, with $R \neq 0$, and put $m_{i,j} = \frac{1}{R}[v_j, v_i]$. Then $m_{i,i} = 0$, $m_{i+1,i} = 1$, and*

$$m_{i,j} m_{i+1,j+1} - m_{i,j+1} m_{i+1,j} = 1$$

for all i and j.

Proof. $m_{i,i} = 0$ because $[u, u] = 0$, and $m_{i+1,i} = 1$ by hypothesis. For any four vectors in the plane the Plücker relation

$$[a, b][c, d] - [a, c][b, d] + [a, d][b, c] = 0$$

holds identically. Taking $(a, b, c, d) = (v_j, v_i, v_{j+1}, v_{i+1})$ and using $[v_i, v_{j+1}] = -[v_{j+1}, v_i]$ gives

$$[v_j, v_i][v_{j+1}, v_{i+1}] - [v_j, v_{i+1}][v_{j+1}, v_i] = [v_j, v_{j+1}][v_i, v_{i+1}] = R^2,$$

and dividing by R^2 is the assertion. □

Theorem 12.3 carries no width. The reductions of Sections 5 and 10 each rest on a continuant window and hold at one width; this holds for all i and j simultaneously.

Theorem 12.4. *Write $v_i = (a_i, b_i)$ and suppose $b_i > 0$ and $b_j > 0$. Then $m_{i,j} > 0$ if and only if $a_j/b_j > a_i/b_i$.*

Proof. $R m_{i,j} = a_j b_i - b_j a_i = b_i b_j \big(a_j/b_j - a_i/b_i\big)$, and $b_i b_j > 0$. □

Positivity of the frieze and clockwise order of the path are therefore one condition, read on the two sides. In particular the positivity of a frieze can be decided on the path, without computing the frieze.

Theorem 12.5. *Fix j. The function $i \mapsto [v_j, v_i]$ satisfies $x_{i+2} = c_i x_{i+1} - x_i$. Consequently a solution of that recurrence agreeing with it at two consecutive indices agrees with it everywhere.*

Proof. $[u, \cdot]$ is linear, so $[v_j, v_{i+2}] = [v_j, c_i v_{i+1} - v_i] = c_i[v_j, v_{i+1}] - [v_j, v_i]$. A solution of a three-term recurrence is determined by two consecutive values. $\square$

The columns of a frieze satisfy that same recurrence, so Theorem 12.5 recovers the path from the frieze: the vertices are the frieze's own columns, and the two constructions are mutually inverse on representatives.

Theorem 12.6. *At width five, with c_i the quiddity of (3), the path satisfies $v_5 = -v_0$ and $v_6 = -v_1$.*

Proof. The recurrence $v_{i+1} = c_i v_i - v_{i-1}$ is the action of $M(c_i) = \left(\begin{smallmatrix} c_i & -1 \\ 1 & 0 \end{smallmatrix}\right)$ on consecutive pairs, and the monodromy of (3) is $M(a_0) \cdots M(a_4) = -I$ by Proposition 3.1. Applying the five factors to (v_0, v_1) returns $(-v_0, -v_1)$. $\square$

The closure is antipodal rather than literal, which is the identification of (a, b) with $(-a, -b)$ used to view $\mathcal{F}_R$ in the plane.

Corollary 12.7. *Let $p \geq 5$ be prime and let (s, t, m) satisfy $stm = p + t + m$. Put $u = ps$, $v = st - 1$ and $g = p(u + v + 1)/uv$, and let $(P, Q) = (gu - p, gv - p)$ be the pair of Theorem 7.7. The path of Definition 12.1 with $v_0 = (1, 0)$, $v_1 = (0, p)$ and c_i the quiddity of (P, Q) is a closed clockwise pentagon in $\mathcal{F}_p$, it is minimal, and distinct triples give pentagons differing by more than a rotation.*

Proof. The vertices are integral and every edge determinant equals $[v_0, v_1] = p$ by Lemma 12.2, so the path lies in $\mathcal{F}_p$; closure is Theorem 12.6; the clockwise order is Theorem 12.4 applied to the positivity of the frieze. For minimality, one edge determinant is p, so the greatest common divisor of the determinants divides p; it equals p only if every $m_{i,j}$ is an integer, that is only if the frieze is integral, which by Theorem 7.7 is not the case for a frieze arising from a solution of the cubic. Distinct triples give distinct rotation orbits of friezes by Theorem 7.7, and the frieze is recovered from the path by Theorem 12.3. $\square$

Corollary 12.7 is checked for every solution of the cubic at each prime $p \leq 31$ by `code/ford_to_pentagon.py` and `code/clockwise_certificate.py`, and the passage back from the pentagon to the frieze by `code/path_to_frieze.py`.

12.1 Surjectivity and the quotient

The construction produces closed clockwise minimal paths. It also produces all of them, and this is not a classification argument: it is forced by the three-vector Plücker identity.

Lemma 12.8. *For any $u, w, x \in \mathbb{Z}^2$, $[w, x]\, u - [u, x]\, w + [u, w]\, x = 0$.*

Proof. Both coordinates expand to zero. $\square$

Theorem 12.9 (The recurrence is forced)**.** *Let u, w, x be three consecutive vertices of a path in $\mathcal{F}_R$, so $[u, w] = [w, x] = R$. Then*

$$R\, x = [u, x]\, w - R\, u.$$

In particular every path in $\mathcal{F}_R$ satisfies the recurrence of Definition 12.1, with $c = [u, x]/R$ the frieze entry, whether or not it was constructed that way.

Proof. Substitute $[u,w]=[w,x]=R$ into Lemma 12.8. $\square$

So the paths of Definition 12.1 are not a special family: a path in $\mathcal{F}_R$ has no freedom beyond its first two vertices, and its coefficients are read off itself.

Theorem 12.10 (Minimality generates the lattice)**.** *Let u, w, u', w' be vertices of a path with $a[u,w]+b[u',w']=1$ for some integers a, b. Then every $x \in \mathbb{Z}^2$ is an integer combination of u, w, u', w', namely*

$$x = a[u,x]\,w - a[w,x]\,u + b[u',x]\,w' - b[w',x]\,u'.$$

Proof. By Lemma 12.8, $[u,w]\,x = [u,x]\,w - [w,x]\,u$ and likewise for u', w'. Multiply the first by a, the second by b and add; the left side is $\big(a[u,w]+b[u',w']\big)x = x$. $\square$

This is what minimality is for. A path whose determinants have greatest common divisor 1 has vertices generating $\mathbb{Z}^2$, so a change of basis between two such paths is integral and not merely rational.

Theorem 12.11 (Transfer)**.** *Let $\gamma=(v_i)$ and $\gamma'=(v'_i)$ be paths with $[v_i,v_j]=[v'_i,v'_j]$ for all i, j, and suppose $a[v_{i_0},v_{j_0}]+b[v_{i_1},v_{j_1}]=1$. Then for every k,*

$$v'_k = a[v_{i_0},v_k]\,v'_{j_0} - a[v_{j_0},v_k]\,v'_{i_0} + b[v_{i_1},v_k]\,v'_{j_1} - b[v_{j_1},v_k]\,v'_{i_1}.$$

Proof. The hypothesis gives $a[v'_{i_0},v'_{j_0}]+b[v'_{i_1},v'_{j_1}]=1$, so Theorem 12.10 applies to γ' with $x=v'_k$. Rewriting each $[v'_\bullet,v'_k]$ as $[v_\bullet,v_k]$ gives the display. $\square$

Theorem 12.12. *If $[v_0,v_1]=R$ then every vertex satisfies $R\,v_i=[v_0,v_i]\,v_1-[v_1,v_i]\,v_0$, that is $v_i = m_{i,0}v_1 - m_{i,1}v_0$.*

Proof. Lemma 12.8 with $(u,w,x)=(v_0,v_1,v_i)$. $\square$

So a path realising a prescribed frieze has no freedom once the initial pair is fixed, and the remaining question is integrality. Writing the two leading columns as $m_{i,0}=\mu_i/N$ and $m_{i,1}=\nu_i/N$, the vertex v_i lies in $\mathbb{Z}^2$ if and only if N divides both components of $\mu_i v_1 - \nu_i v_0$.

The initial pair can be written down. Let $g_0 = \gcd_i \mu_i$ be the greatest common divisor of the numerators in column zero, and set

$$v_0 = \left(\tfrac{g_0}{K},\, 0\right), \qquad v_1 = \left(e,\, \tfrac{N}{g_0}\right), \tag{14}$$

so that $[v_0,v_1]=\frac{g_0}{K}\cdot\frac{N}{g_0}=\frac{N}{K}=R$ as required.

Proposition 12.13. *With the choice* (14)*, the second coordinate of every vertex is an integer.*

Proof. The second coordinate of v_i is $m_{i,0}\cdot N/g_0 = \mu_i/g_0$, an integer because g_0 divides every μ_i. $\square$

So the scale of the initial pair is forced rather than chosen, and only the first coordinate remains, which is the congruence

$$N \mid \mu_i e - \nu_i\,\tfrac{g_0}{K} \qquad \text{for every } i \tag{15}$$

in the single unknown e. That congruence is solvable, and solvable explicitly. Two ingredients are needed.

Lemma 12.14 (Two adjacent columns determine the frieze). *$m_{i,j} = m_{j,0}m_{i,1} - m_{j,1}m_{i,0}$ for all i and j.*

Proof. For fixed j both sides, as functions of i, satisfy the three-term recurrence $x_{i+2} = a_i x_{i+1} - x_i$: the left side because it is a column of the frieze, the right side because it is a fixed linear combination of two columns. At $i = j$ both are 0, and at $i = j+1$ both are 1, the right side because the 2×2 minor of columns 0 and 1 in rows j and $j+1$ equals 1. A solution of a three-term recurrence is determined by two consecutive values. □

Multiplying Lemma 12.14 by N^2 gives

$$\mu_i \nu_j - \nu_i \mu_j = N\, w_{j,i}, \qquad w_{j,i} := N m_{j,i}, \tag{16}$$

and K divides every $w_{j,i}$, being the greatest common divisor of all the numerators.

Theorem 12.15. *Choose integers λ_j with $\sum_j \lambda_j \mu_j = g_0$, and set $e = \frac{1}{K}\sum_j \lambda_j \nu_j$, which is an integer because K divides every ν_j. Then (15) holds for every i. Consequently every positive frieze over $\frac{1}{N}\mathbb{Z}$ of width n and greatest common divisor K arises from a path in $\mathcal{F}_R$, $R = N/K$.*

Proof. Write $d = g_0/K$. Using $Ke = \sum_j \lambda_j \nu_j$ and $Kd = g_0 = \sum_j \lambda_j \mu_j$,

$$K\big(\mu_i e - \nu_i d\big) = \mu_i \sum_j \lambda_j \nu_j - \nu_i \sum_j \lambda_j \mu_j = \sum_j \lambda_j \big(\mu_i \nu_j - \nu_i \mu_j\big) = N \sum_j \lambda_j w_{j,i}$$

by (16). Since K divides every $w_{j,i}$, the sum on the right is K times an integer, so $K(\mu_i e - \nu_i d) = NK \cdot (\text{integer})$. Cancelling K, which is nonzero, gives $N \mid \mu_i e - \nu_i d$. The remaining coordinate is integral by Proposition 12.13, and $[v_0, v_1] = R$ by construction, so (14) is an initial pair realising the frieze. □

Theorem 12.15 is constructive: e is exhibited rather than shown to exist, and the construction is width-uniform. The naive choice g_0/K replaced by 1 does not work, failing for 15 friezes at $N = 12$, which is why the greatest common divisor of column zero enters.

The formula of Theorem 12.11 is pointwise. Read as a function of an arbitrary vector it is a linear map, which is what makes it an element of $SL_2(\mathbb{Z})$ rather than a list of coincidences.

Proposition 12.16. *Fix a, b and four vertices of each path, and define*

$$T(x) = a[v_{i_0}, x]\, v'_{j_0} - a[v_{j_0}, x]\, v'_{i_0} + b[v_{i_1}, x]\, v'_{j_1} - b[v_{j_1}, x]\, v'_{i_1}.$$

Then T is additive and homogeneous, so it is given by an integer matrix; under the hypotheses of Theorem 12.11 it satisfies $T(v_k) = v'_k$ for every k and $[T(v_k), T(v_l)] = [v_k, v_l]$, so its determinant is 1.

Proof. Each $[u, \cdot]$ is linear, so T is a fixed linear combination of linear functions of x, hence linear; the matrix entries are the values at the two standard basis vectors, which are integers. The identity $T(v_k) = v'_k$ is Theorem 12.11. Then $[T(v_k), T(v_l)] = [v'_k, v'_l] = [v_k, v_l]$ by hypothesis, and a linear map multiplying every determinant by its own determinant has determinant 1 once some $[v_k, v_l] \neq 0$. □

So the fibres of the map from paths to friezes are exactly the $SL_2(\mathbb{Z})$ orbits.

Lemma 12.17. *If $A \in SL_2(\mathbb{Z})$ then $[Au, Aw] = [u, w]$.*

Proof. $[Au, Aw] = \det(A)[u, w]$. $\square$

Lemma 12.17 says the frieze is constant on $SL_2(\mathbb{Z})$ orbits of paths, and Theorem 12.11 says a path is determined on its orbit by the frieze together with a minimal path in the class. Together with Theorem 12.9 this settles the correspondence at a prime without appeal to [5]: at $R = p$ and width 5, every non-integral frieze over $\frac{1}{p}\mathbb{Z}$ comes from a solution of the cubic by Theorem 7.7 and hence from a pentagon by Corollary 12.7; every pentagon in $\mathcal{F}_p$ is a path of Definition 12.1 by Theorem 12.9 and so carries a frieze by Theorem 12.3; and two pentagons carrying the same frieze are $SL_2(\mathbb{Z})$-equivalent by Theorem 12.11.

At general R and general width, what remains is the choice of the initial pair (v_0, v_1) realising a given frieze, which is a normalisation and not an identity.

Write $P(R)$ for the number of path classes at width five. Every frieze over $\frac{1}{N}\mathbb{Z}$ has a greatest common divisor dividing N, so $T(N, 5) = \sum_{R|N} P(R)$, and Möbius inversion evaluates P from the counts of Theorem 8.2.

Theorem 12.18. *For every $R \geq 1$ the number of minimal closed clockwise paths of length five in $\mathcal{F}_R$, modulo $SL_2(\mathbb{Z})$, is*

$$P(R) = \sum_{d|R} \mu\left(\frac{R}{d}\right) T(d, 5),$$

with $T(d, 5)$ given by Theorem 8.2. For every prime $p \geq 5$,

$$P(p) = 5\,C(p), \qquad C(p) = \#\{(a, u, v) \in \mathbb{Z}^3_{>0} : auv = p + u + v\}.$$

Proof. The sum over $R \mid N$ partitions the friezes over $\frac{1}{N}\mathbb{Z}$ by greatest common divisor. The class with divisor K is identified with the path classes in $\mathcal{F}_{N/K}$ by Theorems 12.3, 12.4, 12.9, 12.11 and 12.15, so the count depends only on $R = N/K$. Möbius inversion gives the first display. At $R = p$ the divisors are 1 and p, so $P(p) = T(p, 5) - T(1, 5)$, which is $5 + 5C(p) - 5$ by Theorem 7.7 and (2). $\square$

So $\mathcal{F}_p$ carries exactly $5C(p)$ closed clockwise pentagons up to $SL_2(\mathbb{Z})$, and counting them is the problem of Ford; Corollary 12.7 exhibits them. The classification of [5] is cited for context; the correspondence used here is proved above. The values of P are $5, 15, 35, 40, 55, 55, 65, 85, 100, 95$ for $R = 1, \ldots, 10$ (`code/farey_paths.py`).

13 Formalization

The development accompanying this paper is a Lean 4 library against Mathlib. It contains no `sorry`, no `native_decide` and no additional axioms: every declaration cited below depends only on `propext`, `Classical.choice` and `Quot.sound`, which continuous integration checks on each commit.

What is machine-checked is the enumeration and the reductions it rests on. In particular the counts are taken over Definition 2.1 itself rather than over a convenient class: the definition is transcribed with no periodicity assumed (periodicity is derived, not assumed, at widths 4 and 5), and the resulting sets are put in bijection with the arithmetic conditions the search enumerates. Theorem 4.1 and the divisibility statements of Section 6 are verified in that sense.

So is the count at a prime, end to end. Theorem 7.7 and Lemma 9.1 compose in the library into a single declaration giving $T(p, 5) = 5 + 5C(p)$ for every prime $p \geq 5$, stated over Definition 2.1 with periodicity derived rather than assumed, and with $C(p)$ the full solution count of $auv = p + u + v$ rather than a count inside a search box. Theorem 8.2 is verified as well, the support condition and the bijection at each M separately, as are Theorems 8.3, 8.7 and 12.18: the parity of the reflection, the count of the index set by the divisor complement $M \mapsto N^2/M$, and the Möbius inversion are each separate declarations.

The reductions of Sections 10, 11 and 12 are verified in the same sense. Theorems 10.1 and 10.2 are separate declarations, the second in both directions; Theorems 11.2 and 11.3 are stated over an arbitrary commutative domain and over $\mathbb{Z}[i]$ respectively, with the norm bound proved rather than sampled; and Lemma 12.2, Theorems 12.3, 12.4, 12.5 and 12.6 are each a declaration, with the Plücker relation isolated as its own lemma.

The conditions of the classification are formalized as predicates, not paraphrased: minimality as the greatest common divisor of the determinants $[v_i, v_j]$, and the clockwise condition as positivity of those determinants together with positive second coordinates, both stated without division. Lemma 12.8 and Theorems 12.9, 12.10 and 12.11 are separate declarations, as is Lemma 12.17.

What is not machine-checked is stated as such. Theorem 12.18 is proved for all R through Theorem B of [5], which is cited and not reproved. What the present development establishes independently of that citation is the correspondence at a prime and width five, as set out after Lemma 12.17: surjectivity onto paths (Theorem 12.9), the frieze relations (Theorem 12.3), positivity (Theorem 12.4) and injectivity modulo $SL_2(\mathbb{Z})$ (Theorem 12.11). The correspondence is therefore established here for all R and all widths, and no longer rests on Theorem B of [5]: surjectivity onto paths is Theorem 12.9, the frieze relations are Theorem 12.3, positivity is Theorem 12.4, injectivity modulo $SL_2(\mathbb{Z})$ is Theorem 12.11, and the existence of an initial pair is Theorem 12.15. Each ingredient is a Lean declaration; the packaging of the five into a single statement about sets in bijection is assembled here rather than in Lean. Proposition 9.4 is formalized as the slice bijection together with the automatic congruence on the complementary divisor. The external inputs of Shiu and of Linnik are cited, not proved; and the passages to O_ε in Theorem 9.3 and Corollary 7.3 are written proofs. What blocks their formalization is stated precisely rather than left as an absence: Mathlib at the version used here contains no divisor bound of the form $d(n) = O_\varepsilon(n^\varepsilon)$, no Brun–Titchmarsh theorem for multiplicative functions in the sense of Shiu, and no results on divisors in arithmetic progressions. Each of the three would have to be developed before those two statements could be formalized, and none of them is specific to friezes.

A per-statement table naming, for each result, the Lean declaration that verifies it or recording that none exists, is maintained with the library rather than here, since it changes with the library and not with the paper. Tables and computed constants in this paper are regenerated by the scripts named alongside them.

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